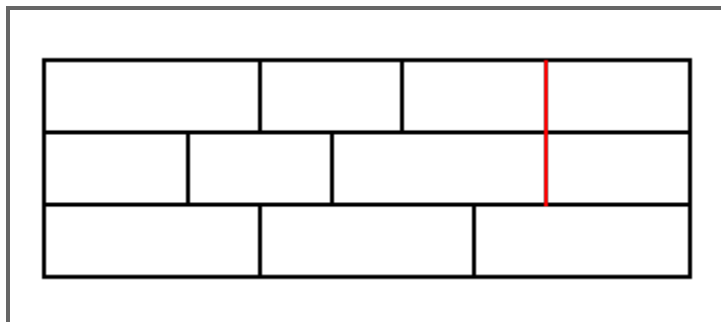


Crack-free Walls

Problem 215

Consider the problem of building a wall out of 2×1 and 3×1 bricks (horizontal $\times$ vertical dimensions) such that, for extra strength, the gaps between horizontally-adjacent bricks never line up in consecutive layers, i.e. never form a "running crack".

For example, the following 9×3 wall is not acceptable due to the running crack shown in red:



There are eight ways of forming a crack-free 9×3 wall, written $W(9, 3) = 8$.

Calculate $W(32, 10)$.